

COMMAND NEXUS · CIT 140 · WEEK 1

Problem Brief

Meridian Forge Robotics · The Process Below

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SECTION 01

Visible Failure

MR-17 suddenly began repeating an old test which is no longer used anymore, test K-44. The operator sent a request telling MR-17 to stop performing the test, however it still continued on. Looking at the evidence it's important to note that MR-17 accepted TEST_K-44_START at 2:13:07 am. The repetition cycle followed at 2:13:09, and when the operator sent a stop request MR-17 returned ACTIVE_INSTRUCTION_PENDING at 2:14:02.

SECTION 02

Verified Facts

The Current scheduling console states that MR-17 does not have an active test assignment.

The Current scheduler also states that test K-44 is not in the current test catalog. This evidence strictly pertains to this current scheduler.

In the Command Telemetry the robotics systems engineer Elias Chen confirmed that the command was validly formatted.

SECTION 03

Assumptions and Open Questions

Nia reported that prior to the incident, MR-17 was functioning normally. Before treating Nia's account as a verified history I would need the record of other staff members who were around during or prior to MR-17 malfunctioning. In the evidence, the staff statements included Nia's report that MR-17 was operating regularly prior to the incident, Elias stating that a controlled restart may clear the active state, and most importantly, the production operations manager Mara Ilyan confirming that production has paused in the affected QA cell.

Open Question : Was every copy, service, or schedule regarding test K-44 fully removed? What exactly did marking the test as retired actually do? The test in question, test K-44, had been marked retired for 186 days prior to the incident. The maintenance archive also states that the test was removed from the active workflow. However, the record fails to confirm whether every copy, service, or schedule for the test was removed.

SECTION 04**System Environment Map****SECTION 04A****People and Roles**

Nia Voss the night QA supervisor, Elias Chen the robotics systems engineer, and Mara Ilyan the production operations manager, all use, support, manage, or are affected by the system. Nia noticed MR-17's actions. Elias Chen confirmed the source label and he has the ability to restart the system if needed. While Mara manages the situation, as they paused production for the affected QA cell.

SECTION 04B**Processes and Rules**

The Current Scheduler controls the incident because it has the ability to assign test assignments.

SECTION 04C**Information and Data**

The incident log, the maintenance record, the command telemetry, and the staff statements matter because they are what can help resolve the situation. They'll hopefully answer the bigger question of who sent the command.

SECTION 04D**Technology and Tools**

Elias described that MR-17 accepted a valid computer command. The source label for the command reads legacy_sync.exe. These facts still don't prove the owner to the service, and the intentions behind the command having run.

SECTION 04E

Environment and Constraints

Mara Ilyan limits time, safety, and the production because following the incident she actually paused production in the impacted QA cell.

SECTION 05

Investigation Boundary

Inside the Investigation

The team will examine MR-17 further to attempt and locate information that will point toward the answer of where exactly the test command was sent from. The location of where the command came from is crucial because it can help prevent the incident from occurring again in the future.

Outside for Now

Current evidence does not yet require a solution because more information is needed to identify the root cause behind the situation having arisen. Only when sufficient information and evidence has been gathered can a solution then be implemented.

SECTION 06

Organizational Impact

Production has slowed largely as a result of the incident because the Production Operations Manager, Mara Ilyan, has paused the production in the affected QA cell. For production to be restored it needs to be running safely again.

SECTION 07

Problem Statement

MR-17 accepted the retired Test K-44 command, therefore it is important to pinpoint why the command was sent and who exactly sent it because production has been paused.

SECTION 08

Next Investigation Step

Gathering information on what the robot's instructions actually accomplish could help answer the question of who actually sent the command. The information that the robot holds right now is important because although a restart may fix its current state, there is no telling whether or not the incident would just happen again in the future. The goal needs to be prevention of the command from happening again.

SHARED ARTIFACT

Working Document Link

https://github.com/joser3500/CIT140-Portfolio/blob/3eef257858e54b334b3e5281be9fab006b1e5c90/Week01_Something_Isnt_Adding_Up/artifact/Romero_Jose%20CIT%20140%20Week%201%20Problem%20Brief.pdf

WEEK 1 STOPPING RULE

Define before designing.

This Problem Brief records the problem that requires investigation. It does not recommend a solution.

